

Hyunseo Lee

heart20021010@gmail.com | (+82)01024884113 | [Github](#) | [LinkedIn](#) | [blog](#)

EDUCATION

Kyungpook National University

B.S. in Computer Science

Daegu, South Korea

03/2021 – Present

- Total GPA: **4.39** / **4.5**, Major GPA **4.45** / **4.5** (credits taken: 134/140)

WORK EXPERIENCE

Research Intern

Visual Intelligence Lab @ETRI

Daejeon, South Korea

01/2026 – 02/2026

- Advisor: **Researcher. Jaeseok Ham**
- Project: Vehicle benchmark dataset construction & baseline model

Research Intern

Visual AI Group @KAIST

Daejeon, South Korea

06/2025 – 08/2025

- Advisor: **Prof. Minhyuk Sung**
- Project: Applying Discrete Flow Matching in Sketch Dataset

Research Intern

PRMI Lab @Kyungpook National University

Daegu, South Korea

03/2023 – Present

- Advisor: **Prof. Woojeoung Nam**
- Thesis: OSCAR: Optical-aware Semantic Control for Aleatoric Refinement in SAR-to-Optical Translation (ICPR 2026)

PUBLICATIONS

International Publications

1. Je-Seok Ham, **Hyunseo Lee**, Yongseon Lee, Kwanyong Park, Jinyoung Moon, Changick Kim[†], "CIVITraj: A Calibration-Free Dataset Construction Framework for Vehicle Trajectory Prediction from Urban CCTV", April. 2026. (under review @Expert Systems with Applications)
2. SangMin Kim, Hokyung Shin, Soeun Bae, **Hyunseo Lee**, Woojeoung Nam[†], "PointRepairFlow: Geometric Offset Denoising for 3D Anomaly Detection", Mar. 2026. (under review @ECCV 2026)
3. **Hyunseo Lee**, Sangmin Kim, Hokyung Shin, Taeheon Kim, Woojeoung Nam[†], "OSCAR: Optical-aware Semantic Control for Aleatoric Refinement in Sar-to-Optical Translation", **ICPR 2026**. [[Paper](#)] [[Code](#)]

Domestic Publications

1. **Hyunseo Lee**, Je-Seok Ham, Jinyoung Moon, "Dataset Construction and Spatial-Map Integration Framework for Infrastructure-Only Vehicle Trajectory Prediction from Urban CCTV", (submitted @ASK 2026)
2. **Hyunseo Lee**, Hee-yeun Ko, Dae-geon Kim, Min-jin Park, Ji-ho Lee, Seok-joo Koh, Du-hyun Koh "Storyboard AI: Design a Diffusion Model-based System and Integrated Pipeline for Dynamic Storyboard Generation", KIIT, Nov. 2025.
3. Minjin Park, Heeyeun Ko, Daegeon Kim, Jiho Lee, **Hyunseo Lee**, Du-hyun Koh, Seok-joo Koh, "The 'Sleeping Worker Problem' and Dynamic Scalability: A Comparative Analysis of AI Deployment Frameworks Flask and FastAPI in Cloud-Native Environments", KIIT, Nov. 2025.
4. Park Jiwon, Yea Changen, Lee Woolyung, **Hyunseo Lee**, "FlowX: Data flow-based GUI editing program for easy data preprocessing", KIIT, June. 2023.

RESEARCH EXPERIENCES

Spatial-Reasoning in VLM with RL

[KNU PRMI Lab](#)

03/2026 – Present

- Improving Spatial Reasoning in VLMs via Supervised Fine-Tuning and Reinforcement Learning with Verifiable Rewards

Inference Time Scaling in Diffusion-, Flow Matching Model

[KNU PRMI Lab](#)

03/2026 – Present

- How to make diffusion, flow matching model reflect reward efficiently and exactly in inference time with limited budget.
- Role: Responsible for all stages of the project under the guidance of a senior researcher.

Dataset Construction for Vehicle Trajectory Prediction from Urban CCTV

[ETRI Visual Intelligence Lab](#)

01/2026 – 02/2026

- Implemented a fully automated pipeline to streamline dataset construction for vehicle trajectory prediction. Furthermore, established a robust baseline model to evaluate the effectiveness of the generated dataset.
- Role: Responsible for all stages of the project under the guidance of a senior researcher.

Implementation of Discrete Flow Matching test bed

[KAIST Visual AI Group](#)

06/2025 – 08/2025

- To analyze the differences between continuous and discrete flow matching, and to distinguish discrete flow matching from autoregressive models, I implemented a testbed from scratch for rigorous verification with sketch dataset.
- Role: Responsible for all stages of the project under the guidance of a senior researcher.
- [\[Github\]](#)

Pansharpening and SAR-to-Optical

[KNU PRMI Lab](#)

03/2025 – 01/2026

- In collaboration with the Korea Aerospace Research Institute (KARI), I implemented SOTA models in pansharpening and SAR-to-optical translation
- Role: Responsible for all stages of the project under the guidance of a senior researcher.
- [\[Project page\]](#)

AWARDS AND HONORS

Academic Excellence Scholarship

08/2021 – 02/2026

- Awarded for all consecutive semesters for maintaining a top-tier GPA.

QUALIFICATIONS

TOEIC: 830

09/2024 – 09/2026

ADsP : Advanced Data Analytics Semi-Professional

03/2024 – Permanent

SQLD : SQL Developer

06/2024 – 06/2026

Linux Master: Lv2

12/2024 – Permanent

TECHNICAL SKILLS

Languages: C, C++, Java, Python, JavaScript

Frameworks: PyTorch, Tensorflow, OpenCV, NextJS

Developer Tools: Git, Docker, Conda, Cursor $\text{\LaTeX}$